# Data Science

## Lecture 09: Association Rule Mining

Prof. Dr. Michael Granitzer

Speaker notes

Converted from LNKB-tex part-en-ds-chp7-association-rule-mining.tex. This is the probability chapter applied to counting: support is a joint probability, confidence a conditional one, and lift a comparison with independence.

# Where We Left Off

Lecture 08 grouped *objects* by similarity. Today groups *items* by co-occurrence — and every measure is a probability from Lecture 02 in disguise.

**Support**
a joint probability

→

**Confidence**
a conditional one

→

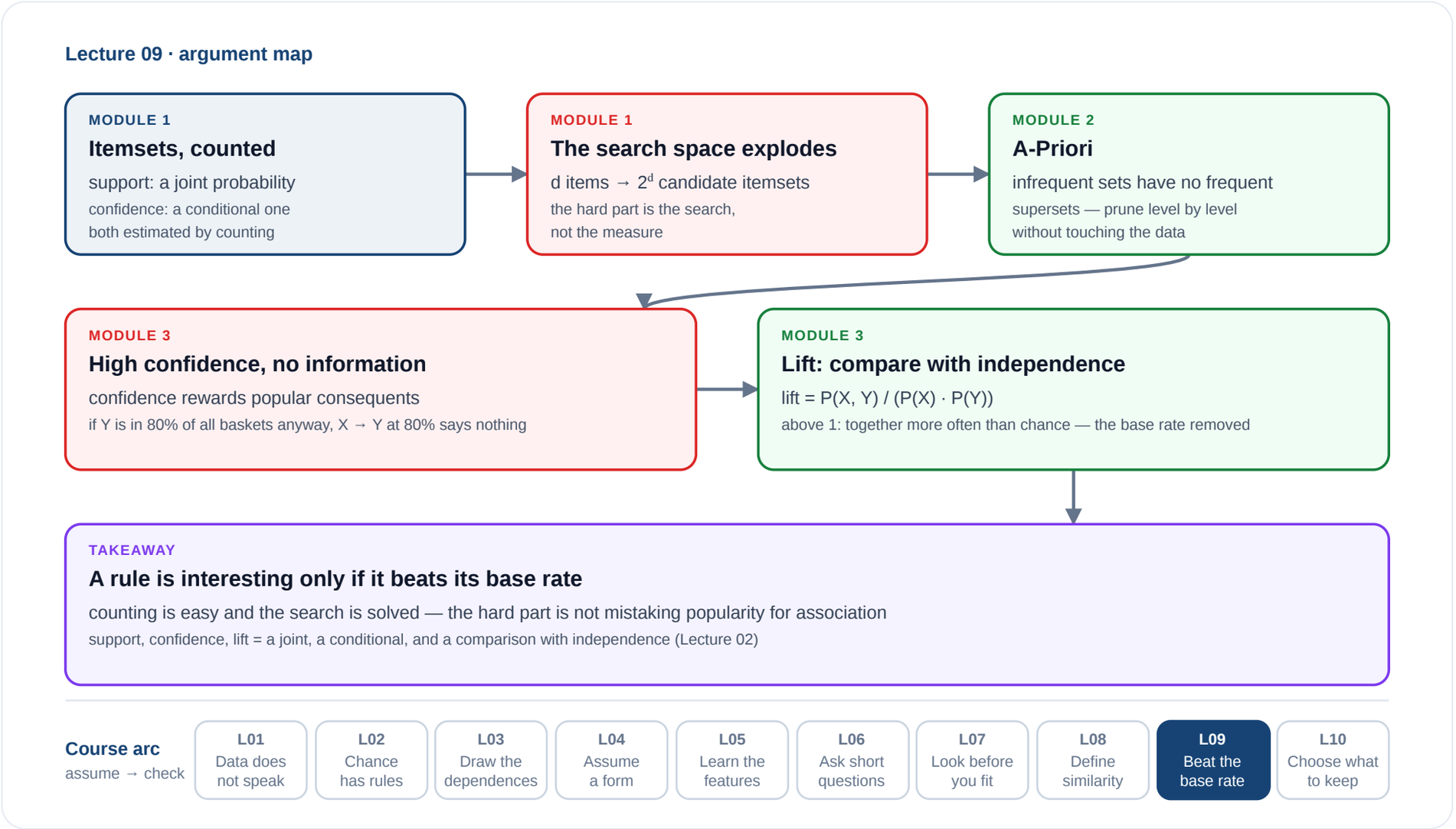
**Lift**
a comparison with independence

> The base-rate trap from Lecture 02 returns: a rule can hold 80% of the time and say nothing.

Speaker notes

Recall the three Lecture 02 objects before any itemset appears: P(A, B), P(B | A) and the independence condition P(A, B) = P(A) P(B). The whole lecture is these three, estimated by counting baskets — plus a search problem that counting alone cannot solve.

# Argument Map



Speaker notes

Counting is easy; the search space is not: $2^d$ candidate itemsets. A-Priori's single structural fact, that infrequent sets have no frequent supersets, prunes most of it without touching the data. Rules are then judged, and confidence alone rewards popular items. Lift divides out the base rate. Point at the strip: "beat the base rate" is the check.

# Module 1

## Itemsets and Their Search Space

> ***Guiding Question:*** *Counting how often items occur together is easy. Why does it become hard with ten thousand products?*

# The Question

Given a set of transactions — shopping baskets, clicked pages, co-occurring symptoms — which **items appear together** more often than chance, and which **imply** which?

### Support

$$P(A \cap B) \geq \sigma_{\min}$$

How often do these items occur in the same transaction?

### Confidence

$$P(B \mid A) \geq \gamma$$

Given $A$ is in the transaction, how often is $B$ too?

Both are quantities the students already know: support is a **joint** probability, confidence a **conditional** one, both estimated by counting transactions.

Speaker notes

Two symbols to keep apart: $\sigma(\cdot)$ is the support *function*, $\sigma_{\min}$ the *threshold* it is compared against. The events on this slide are "the basket contains $A$" and "the basket contains $B$".

Say the connection explicitly — this whole chapter is the probability material applied to a counting problem. Nothing new is being defined, only named for a new context.

# Formalisation

Let $I$ be a set of **items** and $X$ a set of **transactions**, each $x \subseteq I$.

- An **itemset** is any $A \subseteq I$; a $k$**-itemset** has $|A| = k$.
- Its **support** is $\sigma(A) = \frac{|\{x \in X : A \subseteq x\}|}{|X|}$.
- $A$ is **frequent** if $\sigma(A) \geq \sigma_{\min}$.

$\sigma(A \cup B)$ counts baskets that contain **all** items of $A$ **and** of $B$. A union of *itemsets* is an intersection of *events*: $\sigma(A \cup B) = P(A \cap B)$, not "A or B".

# The Search Space

All itemsets over $I$ form a **lattice** ordered by inclusion — with $2^{|I|}$ elements.

A supermarket with 10,000 products has $2^{10000}$ candidate itemsets. Enumeration is impossible; the entire chapter is about avoiding it.

Support and confidence are joint and conditional probabilities estimated by counting. The difficulty is not the measure but the **size of the search space**.

Speaker notes

The lattice picture matters for the next unit: A-Priori prunes whole subtrees of it, and the pruning rule only makes sense once the ordering is clear.

# Module 2

## The A-Priori Algorithm

> ***Guiding Question:*** *What single structural fact about support lets us discard most of an exponential space without looking at the data?*

# The Property That Makes It Possible

> **A-priori property (anti-monotonicity).** Every subset of a frequent itemset is frequent — equivalently: if an itemset is **infrequent, every superset of it is infrequent too**.

Adding an item can only ever *reduce* the number of transactions containing the set. So an infrequent itemset prunes its entire superset subtree from the search.

Speaker notes

The property is obvious once stated and is the whole algorithm. Ask students to justify it in one sentence before showing the consequence — "requiring more items cannot make a set more common".

# The Algorithm

**A-Priori**

$L^1$ = all frequent 1-itemsets.

**for** $k = 2$; $L^{k-1} \neq \emptyset$; $k{+}{+}$: **generate** candidates $C_k = \{a \cup b \mid a \in L^{k-1}, b \in L^1, b \not\subseteq a\}$ **prune** any candidate having an infrequent subset **count** the support of the survivors in one pass over the transactions $L^k = \{c \in C_k \mid \sigma(c) \geq \sigma_{\min}\}$

### Generate

build $k$-candidates from frequent $(k-1)$-itemsets

### Prune

discard candidates with any infrequent subset — no data access needed

### Count

one database pass per level

# Cost and Improvements

### What it costs

One pass over the transactions **per level** $k$ , plus candidate generation. The bottleneck is I/O and the number of candidates.

### How it is improved

Hashing candidates into buckets, transaction reduction (dropping transactions containing no frequent itemset), partitioning and sampling.

A-Priori turns one structural fact — **infrequent sets have no frequent supersets** — into a level-wise search that prunes most of an exponential space without touching the data.

# Mini-Exercise

$\{\text{bread}, \text{milk}\}$ has support $0.18$, and $\sigma_{\min} = 0.22$.

What do you know about $\{\text{bread}, \text{milk}, \text{eggs}\}$ — and how many database passes did it take to know it?

# Mini-Exercise

$\{\text{bread}, \text{milk}\}$ has support $0.18$, and $\sigma_{\min} = 0.22$.

What do you know about $\{\text{bread}, \text{milk}, \text{eggs}\}$ — and how many database passes did it take to know it?

It is **infrequent**: adding eggs cannot raise the support above $0.18$. And it took **no pass at all** — the property alone decides. That is where A-Priori's savings come from.

# Module 3

## From Itemsets to Rules

> ***Guiding Question:*** *A rule holds 80% of the time — and tells you nothing. How is that possible, and what should you measure instead?*

# Generating Rules

An **association rule** $A \Rightarrow B$ (with $A \cap B = \emptyset$) has

$$\text{support} = \sigma(A \cup B), \qquad \text{confidence} = \frac{\sigma(A \cup B)}{\sigma(A)}$$

Every frequent itemset $F$ is split in all possible ways into $A$ and $B = F \setminus A$; rules meeting the confidence threshold are kept.

Support comes from the itemset and is already known. Only confidence must be computed — so rule generation needs **no further database passes**.

Speaker notes

The two-stage structure (find frequent itemsets, then split them) is what makes the whole approach tractable. Confidence is a ratio of two numbers already counted.

# Confidence Is Not Enough

Suppose 80% of all transactions contain milk. A rule $\text{bread} \Rightarrow \text{milk}$ with **confidence 0.80** has learned nothing: milk appears in 80% of baskets regardless of bread.

$$\text{lift}(A \Rightarrow B) = \frac{P(B \mid A)}{P(B)} = \frac{\sigma(A \cup B)}{\sigma(A)\sigma(B)}$$

Lift compares the joint probability against the product of the marginals — the **definition of independence** from Lecture 02, written as a ratio. It is symmetric: $\text{lift}(A \Rightarrow B) = \text{lift}(B \Rightarrow A)$.

Speaker notes

This is the payoff of teaching independence properly. Confidence measures a conditional probability; lift measures the *departure from independence*, which is what "association" was supposed to mean all along. The symmetry follows directly from the second form: swapping A and B changes nothing.

# Reading Lift

| lift | meaning |
|---|---|
| $> 1$ | **positive association**: $A$ and $B$ co-occur more often than independence predicts |
| $= 1$ | **independent**: the rule carries no information |
| $< 1$ | **negative association**: $A$ and $B$ co-occur less often than independence predicts |

Lift is a ratio, not a test. On a rule with low support it rests on a handful of baskets, and a lift of 1.5 can be chance. Before acting, check it for **significance** — the hypothesis tests of Lecture 02 apply.

Speaker notes

Because lift is symmetric, a lift below 1 does not mean "the rule holds in reverse"; it means the two items avoid each other. The significance point is the hook back to hypothesis testing: 20 baskets out of 1,000 are a small sample.

# Mini-Exercise

A retailer reports $\mathrm{nappies} \Rightarrow \mathrm{beer}$ with support 0.02 and confidence 0.60; beer is in 50% of all baskets.

What is the lift — and why is the rule still a weak basis for action?

Speaker notes

Close by connecting all three: support = joint, confidence = conditional, lift = departure from independence. Three probability concepts, one application — and the base-rate trap in a third guise.

# Mini-Exercise

A retailer reports $\text{nappies} \Rightarrow \text{beer}$ with support 0.02 and confidence 0.60; beer is in 50% of all baskets.

What is the lift — and why is the rule still a weak basis for action?

$\text{lift} = 0.60/0.50 = 1.2$: beer is 20% more likely with nappies. But the rule covers only 2% of baskets, so the effect may be chance — test it, and check that it persists.

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> Confidence alone rewards **popular** consequents. **Lift** corrects for the base rate by comparing against independence — the same correction Bayes' theorem makes for priors. Low support calls for a significance check.

# The Churn Shop's Baskets

The shop delivers grocery boxes. Of **100** boxes, **80** contain milk, **20** contain cereal, **16** contain both.

The rule $\mathrm{cereal} \Rightarrow \mathrm{milk}$ holds with confidence 80%. Is it worth a "customers who order cereal also order milk" banner?

Speaker notes

The running example once more, now as baskets. Lift 1 means P(milk | cereal) = P(milk): the two items are independent in this data, and the 80% was the popularity of milk. The banner would change nothing. Contrast with the exercise before: lift 1.2 was a real, if small, association. The check of the course arc, in its plainest form: compare with what chance alone predicts.

# The Churn Shop's Baskets

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$$\text{conf} = \frac{16}{20} = 0.8 \qquad \text{lift} = \frac{0.8}{0.8} = 1.0$$

No: milk is in 80% of **all** boxes. Cereal buyers take milk exactly as often as everyone else — the rule restates the base rate.